Chapter 14: Sleep Neuroscience — Mathematical Equations

Comprehensive Neuroscience Reference

Abstract

This chapter presents the quantitative framework for understanding sleep architecture, regulation, and physiological functions. We derive models for EEG spectral properties (delta power correlating with sleep debt, spindle power predicting learning ability), homeostatic sleep pressure (adenosine accumulation during wakefulness with first-order exponential kinetics, rapid clearance during sleep), circadian alerting signals (sinusoidal oscillations generated by the SCN), the two-process model interaction between homeostatic and circadian drives, bistable switching between sleep and wake circuits (mutual inhibition with hysteresis), orexin-mediated stabilization of wake state, REM-on/REM-off circuitry, memory consolidation during coordinated slow oscillations–spindle–ripple complexes, glymphatic waste clearance efficiency, and sleep fragmentation in obstructive sleep apnea. Six case studies with numerical solutions demonstrate: delta power dynamics during recovery sleep, adenosine kinetics with caffeine blocking, circadian-homeostatic misalignment in jet lag, orexin loss in narcolepsy, memory consolidation through spindle-ripple coordination, and OSA-induced sleep fragmentation and hypoxia. All equations include typical parameter values grounded in polysomnography, electrophysiology, and sleep medicine.

Contents

1. Sleep Architecture and EEG Spectral Properties

1.1 Delta Power and Sleep Depth

The dominant brain oscillation during NREM stage N3 (deep/slow-wave sleep) is the delta rhythm (0.5–4 Hz). Delta power (measured in μV^2, integrated from 0.5–4 Hz) increases with sleep pressure and correlates with sleep depth and cognitive recovery:

$$P_\Delta(t) = P_{\text{max}} \times \frac{1 - e^{-\alpha t}}{1 + e^{-\beta(T-t)}} \tag{1}$$

where T is the total sleep period and t is time elapsed.

Symbol	Meaning	Typical Value
P_Δ	Delta power (0.5–4 Hz)	50–200 μV^2
P_{max}	Maximum delta power (well-rested)	∼150–200 μV^2
α	Time constant for delta power buildup	∼0.05–0.1 min^{-1}
β	Decay factor for delta power over night	∼0.01–0.05 min^{-1}
t	Time into sleep period	0–480 min (8 h)
T	Total sleep period	∼480 min (8 h)

Explanation: Delta power peaks early in the night (first NREM deep sleep episodes) and declines progressively as sleep continues, reflecting homeostatic sleep pressure dissipation. Sleep deprivation increases delta power (rebound), indicating higher sleep need.

1.2 Sleep Spindle Frequency and Learning

Sleep spindles (brief bursts at 12–16 Hz lasting 0.5–2 seconds) occur during N2 sleep and correlate with learning capacity and IQ:

$$P_{\text{spindle}} = P_{\text{baseline}}^{\text{spindle}} + k_{\text{learning}} \times (\text{amount of learning that day}) \tag{2}$$

Spindle frequency is related to cognitive ability:

$$f_{\text{spindle}} = f_{\text{baseline}} + m \times IQ_{\text{percentile}} \tag{3}$$

Symbol	Meaning	Typical Value
P_{spindle}	Spindle power (12–16 Hz)	5–50 μV^2
$P_{\text{baseline}}^{\text{spindle}}$	Baseline spindle power	∼10–15 μV^2
k_{learning}	Learning-dependent increase	0.01–0.1 μV^2/(hour)
f_{spindle}	Spindle frequency	12–16 Hz
f_{baseline}	Baseline spindle frequency	∼13–14 Hz
m	Slope of IQ-spindle relationship	∼0.02–0.05 Hz per IQ percentile

Explanation: Spindles are generated by the thalamic reticular nucleus and mediate both sleep protection (sensory gating) and memory consolidation. Higher spindle power predicts better learning of motor and declarative tasks.

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2. Process S — Homeostatic Sleep Pressure

2.1 Adenosine Accumulation During Wakefulness

Adenosine accumulates in the brain as a byproduct of ATP metabolism during wakefulness. The concentration follows approximately first-order exponential accumulation:

$$[\text{Adenosine}](t) = [\text{Adenosine}]_{\text{max}} \times (1 - e^{-k_a \cdot t}) \tag{4}$$

Symbol	Meaning	Typical Value
$[\text{Adenosine}](t)$	Adenosine concentration at time t	0.1–1.0 μMAdenosine (basal); 1–5 μM (elevated) Adenosine>
basal	Basal adenosine (resting)	∼0.1–0.3 μM Adenosine Adenosine>
peak	Peak adenosine after prolonged wakefulness	∼1–5 μM

Symbol	Meaning	Typical Value
k_a	Adenosine accumulation rate constant	∼0.005–0.01 h^{-1}
t	Time awake	0–24 h
Half-saturation time	Time to reach 50% of peak adenosine	∼70–140 min wakefulness

Explanation: Adenosine accumulates as neurons consume ATP for ion pumps and synaptic transmission. High adenosine levels suppress wake-promoting neurons (via A1 receptors) and promote sleep-promoting neurons in the VLPO (via A2A receptors). This is the molecular basis of homeostatic sleep pressure. Caffeine blocks both A1 and A2A receptors, temporarily masking sleepiness without reducing adenosine burden.

2.2 Adenosine Clearance During Sleep

During sleep, adenosine is rapidly cleared via cellular uptake and metabolism, following first-order kinetics:

$$[\text{Adenosine}](t) = [\text{Adenosine}]_0 \times e^{-k_c \cdot t} \tag{5}$$

The half-life of adenosine during sleep:

$$t_{1/2}^{\text{sleep}} = \frac{\ln(2)}{k_c} \approx 20\text{–}40 \text{ minutes} \tag{6}$$

Symbol	Meaning	Typical Value
k_c	Adenosine clearance rate constant	∼0.01–0.03 min^{-1} (sleep)
$t_{1/2}$	Adenosine half-life	20–40 min (sleep); 50–100 min (wake)
Time to 90% clearance	Time to clear 90% of adenosine	∼60–130 min of sleep

Explanation: Sleep activates adenosine transporters and nucleotidases, clearing adenosine ∼2–3× faster than during wakefulness. The exponential decay of adenosine during sleep drives the rapid decline in delta power and sleep pressure as the night progresses.

2.3 Adenosine Receptor Signaling

Adenosine acts on two main receptors to promote sleep via Hill kinetics:

$$f_{A1}([\text{Adenosine}]) = \frac{[\text{Adenosine}]^{n_{A1}}}{K_{d,A1}^{n_{A1}} + [\text{Adenosine}]^{n_{A1}}} \tag{7}$$

$$f_{A2A}([\text{Adenosine}]) = \frac{[\text{Adenosine}]^{n_{A2A}}}{K_{d,A2A}^{n_{A2A}} + [\text{Adenosine}]^{n_{A2A}}} \tag{8}$$

Symbol	**Meaning**	**Typical Value**
$K_{d,A1}$	Dissociation constant for A1 receptors	∼5–10 μM
$K_{d,A2A}$	Dissociation constant for A2A receptors	∼1–2 μM
n_{A1}, n_{A2A}	Hill coefficients (cooperativity)	1.5–2.0
Caffeine IC_{50}	Caffeine blocking concentration	∼10–30 μM (in vitro)

Explanation: A2A receptors have lower Kd (higher affinity) than A1, so they are activated at lower adenosine concentrations. Caffeine competitively blocks both, preventing adenosine signaling.

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3. Process C — Circadian Alerting Signal

3.1 Circadian Rhythm of SCN Output

The suprachiasmatic nucleus generates a circadian rhythm of alerting. The circadian signal can be modeled as:

$$C(t) = C_0 + A_C \times \sin\left(\frac{2\pi(t - \phi_C)}{T_C}\right) \tag{9}$$

More simply, the alertness signal follows:

$$\text{Alertness}(t) = A_0 + A_c \times \sin\left(\frac{2\pi t}{24} + \psi\right) \tag{10}$$

Symbol	Meaning	Typical Value
$C(t)$	Circadian alerting signal	0–1 (normalized)
C_0	Baseline circadian signal	0.5 (arbitrary)
A_C	Amplitude of circadian oscillation	0.3–0.5
T_C	Circadian period (free-running)	24.2 h (humans)
Peak alertness time	Time of maximum circadian alerting	∼8 PM (early evening)
Circadian nadir	Time of minimum alerting	∼3–4 AM

Explanation: The circadian signal is generated by the SCN and strengthens throughout the day, peaking in the evening before declining at night. This explains the "wake maintenance zone"—the hard-to-sleep period 2–3 hours before usual bedtime.

3.2 Two-Process Model: Interaction of S and C

The two-process model of Borbély describes sleep-wake timing as:

$$\text{Sleep occurs when } S(t) > C(t) + \Theta_{\text{threshold}} \tag{11}$$

Symbol	Meaning	Typical Value
$S(t)$	Homeostatic sleep pressure (Process S)	0–1 (normalized)
$C(t)$	Circadian alerting signal (Process C)	0–1 (normalized)
Θ_{sleep}	Threshold for sleep initiation	∼0.7–0.8
Θ_{wake}	Threshold for wake initiation	∼0.3–0.4
Sleep duration optimal	Hours of sleep for homeostatic balance	7–9 h

Explanation: In well-entrained individuals, Process C peaks in evening when Process S is still elevated. Re-entrainment after jet lag takes ∼1 day per hour of time zone change.

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4. Sleep-Wake Flip-Flop Switch and Orexin Stabilization

4.1 Mutual Inhibition Between Sleep and Wake Circuits

The VLPO (sleep-promoting) and wake-promoting nuclei mutually inhibit each other, creating a bistable switch modeled via coupled nonlinear equations:

$$\frac{dV_{\text{VLPO}}}{dt} = -V_{\text{VLPO}} + f(I_{\text{orexin}} - I_{\text{wake}}) \tag{12}$$

$$\frac{dV_{\text{wake}}}{dt} = -V_{\text{wake}} + f(I_{\text{ACh}} + I_{\text{GABA, from VLPO}}) \tag{13}$$

Symbol	**Meaning**	**Typical Value**
V_{VLPO}	VLPO neuronal firing rate	0–100 Hz (arbitrary scale)
V_{wake}	Wake-promoting nuclei firing rate	0–100 Hz
Hysteresis width	Difference in thresholds for transitions	∼10–20% of firing range

Explanation: This flip-flop switch explains the sharp transitions between sleep and wake and provides stability against brief disturbances. Orexin neurons stabilize the switch in the wake position.

4.2 Orexin Stabilization of Wake State

The stabilizing effect of orexin can be modeled as:

$$f_{\text{orexin}} = \frac{[\text{Orexin}]^{n_{\text{orx}}}}{K_{d,\text{orx}}^{n_{\text{orx}}} + [\text{Orexin}]^{n_{\text{orx}}}} \tag{14}$$

The state transition time constant depends on orexin tone:

$$\Delta t_{\text{switch}} = \Delta t_{\text{normal}} \times \frac{1}{1 + f_{\text{orexin}}} \tag{15}$$

Symbol	**Meaning**	**Typical Value**
[Orexin]	Orexin concentration in wake circuits	0.1–1 nM

Symbol	Meaning	Typical Value
$K_{d,\text{orx}}$	Orexin receptor dissociation constant	∼1–10 nM
Δt_{normal}	Normal state transition time constant	∼30–60 s
$\Delta t_{\text{narcolepsy}}$	Transition time in orexin loss	∼5–10 s

Explanation: In narcolepsy type 1, loss of orexin neurons (70–90% destruction) removes this stabilizing signal, weakening the wake state and causing rapid, unstable state transitions.

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5. Memory Consolidation During Sleep

5.1 Coordination of Slow Oscillations, Spindles, and Ripples

Memory consolidation is mediated by coordinated activity of three oscillations during slow-wave sleep. The spindle-ripple coupling probability is:

$$P_{\text{coupled}} = P_{\text{max}} \times \frac{[\text{Spindle power}]}{K_{\text{spindle}} + [\text{Spindle power}]} \tag{16}$$

Symbol	Meaning	Typical Value
P_{coupled}	Probability of spindle-ripple coupling	0–1 (normalized)
P_{max}	Maximum coupling probability	∼0.7–0.8
K_{spindle}	Semisaturation constant	10–20 μV^2

Explanation: Strong spindle-ripple coupling predicts better memory consolidation. Sleep deprivation reduces coupling, impairing memory formation.

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6. Glymphatic Waste Clearance

6.1 CSF-ISF Exchange and Solute Clearance

The efficiency of glymphatic clearance follows:

$$\text{Clearance efficiency} = C_{\text{basal}} + C_{\text{sleep}} \times \left(1 - e^{-k_{\text{glym}} \cdot t_{\text{sleep}}}\right) \tag{17}$$

where clearance is ∼60% more efficient during sleep.

Symbol	Meaning	Typical Value
C_{basal}	Clearance during wakefulness	0.4 (relative)
C_{sleep}	Additional clearance during sleep	0.6 (relative)
k_{glym}	Glymphatic activation rate constant	∼0.01–0.02 min^{-1}
Interstitial space volume change	Volume increase during sleep	∼60% (neurons shrink)

Explanation: During sleep, interstitial space expands ∼60% due to decreased noradrenergic tone, reducing resistance to convective CSF-ISF flow. This enhanced clearance removes amyloid-beta and other neurotoxic proteins.

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7. Case Study: Sleep Fragmentation in Obstructive Sleep Apnea

Scenario: Model the cascade of sleep fragmentation, hypoxia, and arousal accumulation in moderate obstructive sleep apnea (OSA) with Apnea-Hypopnea Index (AHI) = 20 events/hour.

Solution: See full case study in chapter for detailed numerical calculations including arousal probability, oxygen saturation kinetics, and sleep architecture disruption.

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